Macromolecular simulation software workshop

October 12, 2015 to October 25, 2015

Location : CECAM-DE-JUELICH

Introduction

As the use of biomolecular simulation grows, a corresponding boom in (decentralised) software development is taking place. Much of this is within multidisciplinary research groups and is highly focussed on the specific needs of that community. Development is often done by researchers with little or no formal training in software engineering or programming. This is not entirely a negative – it encourages practical solutions to real-life problems and 'thinking outside the box'. However, it is obvious that it is not the ideal route to the production and maintenance of high quality, flexible, sustainable, and usable software products that can be adopted and modified by others. Giving researchers the right tools and training to develop such software will reduce the amount of 'wheel-reinvention', and consequently improve research productivity.

This CECAM Software Development Workshop gave an opportunity for representatives of these projects and many others that are as yet less well-known, to interact with end users with the aim of tackling the most challenging problems in this domain, particularly those relating to how the capabilities of and opportunities presented by future generations of massive, sometimes distributed, heterogeneous, computational resources can best be leveraged in this domain of science.

This highly interactive event, targeted principally at early stage researchers, was formatted as six sequential workshops, each 2 days long. Participants were not required to sign up for the full duration of the course, but typically attended a sequence of one or more sub-workshops over the period that particularly interested them or addressed their needs. Each sub-workshop had 35-40 participants, and overall 93 researchers took part at some stage, including 15 from the USA, through additional support provided by the NSF.

Below, each of the sub-workshop leads comments on the activity they organised.

1. **''Software carpentry workshop'' (instructors: Philip Fowler, Oxford and David Dotson, ASU) Mon 12/10-Tue 13/10**

We ran a Software Carpentry workshop covering shell scripting, python and version control using git to enable participants who were less experienced to learn these tools which were pre-requisites for the remainder of the CECAM workshop. These workshops are interactive with one instructor live coding on the screen, whilst the other roams the room solving problems. An anonymous questionnaire was collected at the end of the workshop; all learners agreed with the statement "I enjoyed the workshop". For more details, including more feedback please visit this site.

2. **"Analysing simulation data" (lead: Philip Fowler, Oxford) Wed 14/10-Thu 15/10**

The current practice in analysing simulation data involves either "black-box" compiled C programs (such as the g_tools that are part of GROMACS) or limited scripting capabilities built-in to molecular viewers (such as Tcl in VMD). A shift is coming to the use of high-level languages, such as python. This has three key advantages: (i) development is open and the logic of an underlying algorithm can be easily understood, (ii) an analysis tool can read trajectories from any simulation code and (iii) the researcher has access to the ecology of powerful python modules, enabling analysis that otherwise would have been very difficult. The developers of two such python modules, MDAnalysis and pmx, were invited to this workshop. Each introduced either module with a short talk, followed by an extensive hands-on tutorial. At the end of the first day, the participants proposed research questions that could be solved using these tools and six teams formed. The following day, the six teams worked on their projects, under guidance from the assembled experts and academic software developers before presenting back to the group (i.e. a HackDay). From feedback, all the participants learnt something useful that will help their research and would also recommend a workshop like this one to other researchers, whilst 88% felt confident enough to now contribute to an academic open source software project. The biggest complaint by far from the participants was the poor Wifi. This made installing software and running a collaborative HackDay very difficult. For example, two teams decided to work in the onsite cafeteria on the second day was the Wifi was better. For more information please see this site.

3. **"Setting up simulations" (lead: Charlie Laughton, Nottingham) Fri 16/10-Sat 17/10**

On the basis of "rubbish in, rubbish out", it is vital to pay as much attention to the preparation of systems for molecular dynamics simulation as it is to work on optimising the performance of the MD codes themselves and methods to analyse the data they generate. Automated, standardised, set-up tools offer opportunities for enhancing quality (e.g. by automatically identifying errors in input structures), reproducibility (e.g. by creating shareable, well-documented, workflows), and usability (e.g. in high-throughput simulation campaigns).

The aim of this sub-workshop was to introduce participants to alternative paradigms and toolkits for setting up molecular dynamics simulations, including validating starting

structures and repairing "errors", embedding solutes in solvents and membranes, relaxation and equilibration protocols, and setting up complex simulations such as free energy perturbation calculations. Participants had the opportunity to explore the capabilities of a range of web-based (e.g. MDWeb) and python toolkit-based solutions (e.g. FESetup, MemProtMD), and in a "hackathon" type activity, worked in small groups with the code owners to tackle problems related to unmet needs, flexibility, interoperability, and performance.

The enthusiasm and hard work of the participants made this a very successful event. Many participants were introduced to possibilities of ways of working that they had never had a chance to consider before. During the “Hackathon” day attendees from widely differing backgrounds had an opportunity to learn how to work together effectively in small teams to develop software to address genuine and pressing needs. More information is available on the course website.

4. **“Developing interoperable and portable molecular simulation software libraries” (lead: Julien Michel, Edinburgh) Mon 19/10-Tue 20/10**

We ran a workshop on best practices to efficiently write molecular simulation software that will be reusable by others. Over the past few years there has been a trend towards writing molecular simulation libraries rather than new monolithic piece of softwares. This has the advantage that i) code can be more easily shared and re-used in other projects ii) new developers have access to communities of fellow developers. However fully embracing this model requires changes in current software engineering practices in the field. During the first session the invited speakers discussed their perspective on software engineering for molecular simulation projects. This was followed by a group discussion. The main findings were that most code developers have a background i chemistry or physics and need to be explicitly taught the benefits of software engineering early in their studies. One major benefit is that developers should write well documented and portable code for their ‘future-selves’, and that the long term productivity gains compensate for the perceived additional burden in documenting code and writing unit tests. This was followed by a hands-on tutorial on how to develop with the Plumed molecular simulation library. The tutorial was unfortunately severely hampered by very poor wireless networking. On the second day the delegates followed an hands-on tutorial on the molecular simulation library OpenMM. This was followed by a ‘free-form’ session where delegates broke-out in three subgroups that were free to discuss topics of their interest with the invited speakers. Topics for the free-form session had been collected prior to the meeting and reviewed by the invited speakers/developers for feasibility/appropriateness. From the feedback received all delegates were satisfied/very satisfied with the workshop. The biggest complaints were related to the difficulties of using the Wifi on-site, and the difficulties of traveling to the venue.
Results of a feedback form are available here

5. **“High Performance Distributed Computing tools for Biomolecular Simulations" (lead: Shantenu Jha, Rutgers) Wed 21/10-Thu 22/10**

The objective of Workshop 5 was to introduce participants to tools that would enable them to utilize high-performance computing towards their science. There were two main parts of Workshop 5; the first part introduced students to the suite of RADICAL-Cybertools [1], the second to Copernicus [2].

Workshop 5 assumed there would be a wide diversity in the background and interests of students/participants. This would range from the “expert end-user” to the “disinterested black-box” user. Indeed, our pre-workshop survey confirmed the broad range of skills and expectations (“I just want to use this tool” to “Its interesting to know how this works under the hood”). Ultimately managing the expectation of a wide range of participants was the greatest challenge.

In anticipation, we designed Workshop 5 to have several tracks. Furthermore, In order to provide the students with hands on experience, we arranged for access to US NSF funded supercomputers: Stampede and Comet. This required a (separate) proposal for computing time (100,000 SUs), working with the XSEDE workflows team to provide an “access point” for all participants so they could initiate their work onto XSEDE.

The RADICAL-Cybertools part [3] in turn was comprised of three components: the first was an introduction to RADICAL-Pilot [4] the second a quick introduction to Replica-Exchange framework [5] and the third component an overview of Ensemble-MD toolkit [6]. The RADICAL-Pilot and the Ensemble-MD framework component in turn had three main tracks. The first was an introduction to the tool and a “walk through” a series of exercise to help understand these tools; the second was to use the tool on a meaningful problem at meaningful scale; the third track was to use the tool in an “open” exercise. The Replica-Exchange framework only had the first two parts.

In the open exercises, students took the tools they had learnt and were encouraged to apply them to their own problems. Participant responses to the open exercises were mostly along the following lines: (i) “These tools are great and we’d like to use them on our home machine, can you help us install/provide them?”, (ii) “These tools will be useful if I can use it with this application, or modify to do X”. Students started to think of tools/software as a way to develop “composable methods”. Limited time meant that many of the “extensions” did not reach completion, but some started the process.

The participants learnt the importance and power of “advanced” that enable them to use resources more efficiently. An important consideration for all scientists is how to determine which tools to use and how much effort to invest in making “buggy(!), developing, unpolished” tools work. These were some considerations that were discussed and students were asked to respond to. [8,9]

An important lesson for the organizers of the Workshop 5 and RADICAL-Cybertools component in particular, was the significant amount of set-up to get the exercises going. This was mainly because we wanted to give students the full experience of running jobs on supercomputers. This in turn is necessary if we students are to learn the advantage of "scalable computing" as opposed to "just doing it on your laptop". If we had limited our tutorial to a laptop or a virtual machine, many of these problems would have been localized and controlled. The desire to give students an "in vivo" experience on production supercomputer/cyberinfrastructure is a non-trivial challenge.

In general, The setup of a controlled environment which would allow students to perform a set of sophisticated tasks is a non-trivial exercise. This required significant perturbation to the default software environment of "workflows.iu.xsede.org" machine (which served as the starting/access point).

1. http://radical-cybertools.org/
2. https://github.com/kassonlab/copernicus.git
3. https://github.com/radical-cybertools/tutorials/wiki/CECAM-2015/
4. https://radicalpilot.readthedocs.org/
5. http://repex.readthedocs.org/
6. http://radicalensemblemd.readthedocs.org/
7. https://drive.google.com/file/d/0B4XS8WEUG0G4UW9tOEZGaVlzUVE/
8. https://docs.google.com/spreadsheets/d/1JVc-iWjWm3Pwd2ESZq-Zu0YxRq-HxaY3VPLCK-Nv8wA/
9. https://docs.google.com/spreadsheets/d/1bw1XqRQ7wqWiHvhkhVFkwI1dSbRKVuMEeIj6pmfSwBI/

6. **"Advanced Sampling and Long Timescale Molecular Dynamics" (lead: Cecilia Clementi, Rice) Fri 23/10-Sat 24/10**

The final workshop focused on software tools for advanced sampling methods. The first day covered tools developed by the ExTASY project (www.extasy-project.org) A short lecture presented the Diffusion Map-directed-MD method (based on Locally Scaled Diffusion Maps) and COCO (based on Principal Component Analysis), then the remainder of the day was devoted to guided tutorial exercises. The DM-d-MD and COCO methods are implemented as workflows built using the RADICAL EnsembleMD package, presented in workshop 5, and so the software environment was very familiar to delegates. In the first tutorial session we walked through how to run two pre-defined workflows on remote HPC resources, this was followed by a walk-through the process of implementing a new workflow with a different combination of tools, and in the final session, delegates worked on either sampling a small peptide using COCO or on modifying/extending the example workflows for their own use. We gathered qualitative

feedback from delegates, which were generally positive, although they found that the software was quite ‘bleeding edge’ and that they would have liked more time to experiment with the individual tools as well as running the workflows. One commented that they were left a bit confused by the wide array of tools (and levels of abstraction) layered on top of RADICAL-Pilot over the previous 3 days.

The second day was taken up with a PyEMMA workshop (ca. 35 participants) Presented by Prof. Dr. Frank Noé (FU Berlin), Dr. Guillermo Pérez-Hernández (FU Berlin), and Dr. Antonia Mey (Univ. Edinburgh). This workshop was an introduction to the Markov modeling package PyEMMA, an open-source (LGPL) Python package for analyzing molecular dynamics simulations in such a way that models of the thermodynamics and kinetics of potentially complex molecular processes can be obtained, analyzed, interpreted and compared to experimental data.

After a scene-setting introduction, the workshop consisted of interactive hands-on tutorials where the participants were asked to complete certain tasks after having seen a simple example or part of the solution while using the PyEMMA online documention. At the end of each part, the solution was presented by the workshop organizers.

The workshop had significant impact by introducing a selected group of interested participants to the PyEMMA software. These participants are likely to keep using the software and propagate their knowledge further by introducing it to other potential users in their working groups. Some feedback was immediately received by the fact that several participants became watchers or stargazers of the PyEMMA project on github.

Summary:

This new format of event for CECAM has clearly been a great success. Feedback from participants has been extremely positive. Over ninety early-stage researchers, from across Europe and the US, have had an opportunity to not just learn about, but to contribute to, cutting-edge innovations in molecular simulation software and methodology. In addition the format promoted networking and group-working opportunities and skills that will help promote a secure and exciting future for the next generation of molecular software developers and users.

For the perspective of the workshop teams, there were also very useful pedagogical lessons learnt for future activities.It was very interesting to compare the different approaches to running different workshops (e.g. synchronous formats where all participants worked through an example problem step-by-step together, versus asynchronous formats where participants worked at their own pace, supported by circulating experts), and also alternative approaches to delivering software to the heterogenous computing resources that participants brought with them (direct installation of packages, virtual machines, Dockers, etc.).

In addition to the teams presenting and supporting the individual workshops, we would like to thank the administration team at Forschungzentrum Juelich for their vital assistance with the logistics of this complex event.

Iain Bethune
Paolo Carloni
Cecilia Clementi
Phil Fowler
Shantenu Jha
Charlie Laughton
Julien Michell

4/12/15